

1202SCG Calculus 1
1020ENG Engineering Mathematics 2

Week 1 Lecture

David Harman

Teaching Team

Convenor: Jason Hay
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Lecturer: David Harman

There will also be workshop leaders, PASS leaders, and markers assisting with the running of the course.

Assessment Overview

- ▶ **20%:** Weekly Workshops (Starting in Week 2)
- ▶ **40%:** Mid-Trimester Exam
- ▶ **40%:** End of Trimester Exam

There is a **Mandatory Pass Component** on the end of trimester exam. A minimum of 40% on the final exam is required to be eligible for a passing grade at the end of the trimester.

Assumed Knowledge

Mathematical Methods

OR

1017SCG Foundation Mathematics

Content Overview

- ▶ Revision of Functions
- ▶ Limits and Continuity
- ▶ Introduction to Derivatives
- ▶ Rules for Differentiation
- ▶ Turning Points, Concavity, and Applications of Derivatives
- ▶ L'Hopital's Rule and Taylor Series
- ▶ Basics of Integration
- ▶ Further Techniques of Integration
- ▶ Introductory Differential Equations
- ▶ Further ODEs with Applications

Revision of Functions

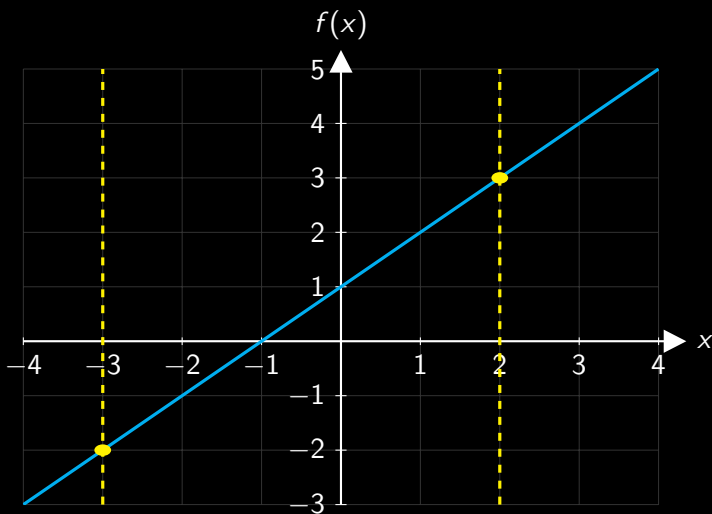


Functions can only have one output.

Functions must be consistent.

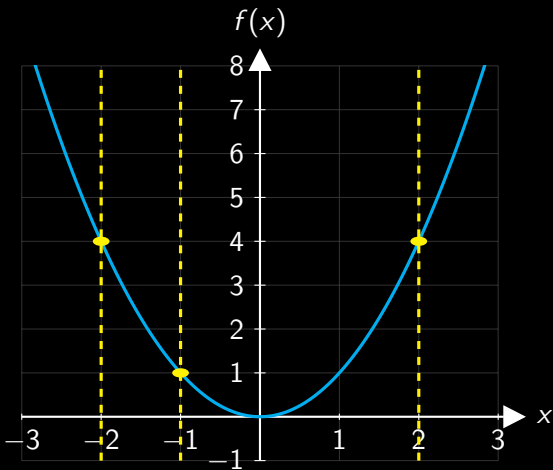
Vertical Line Test

$$f(x) = x + 1$$



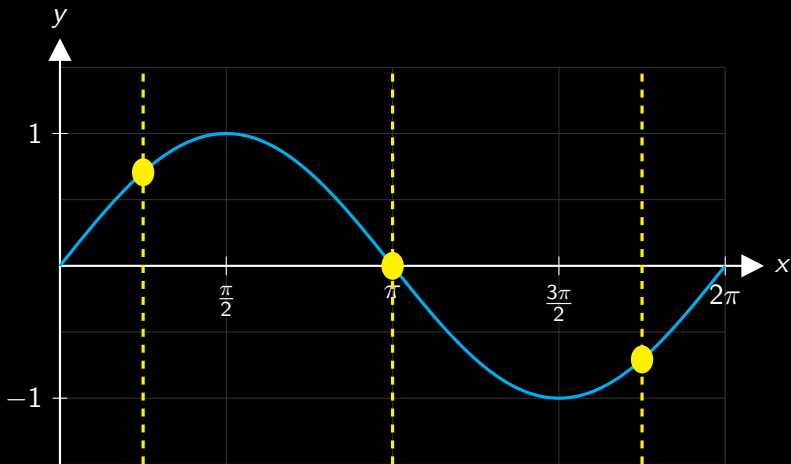
Vertical Line Test

$$f(x) = x^2$$

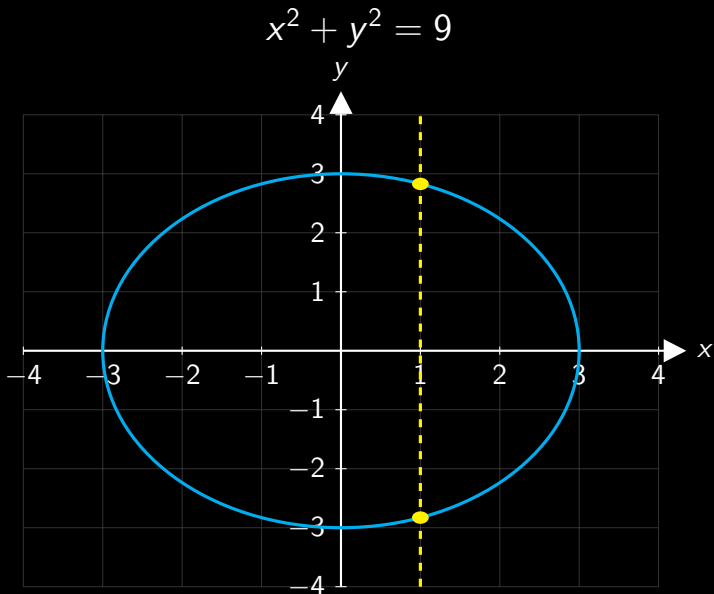


Vertical Line Test

$$f(x) = \sin(x), \quad 0 \leq x \leq 2\pi$$

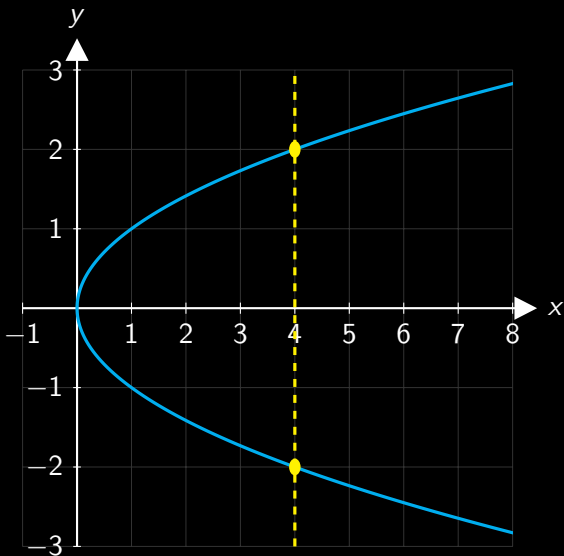


Vertical Line Test

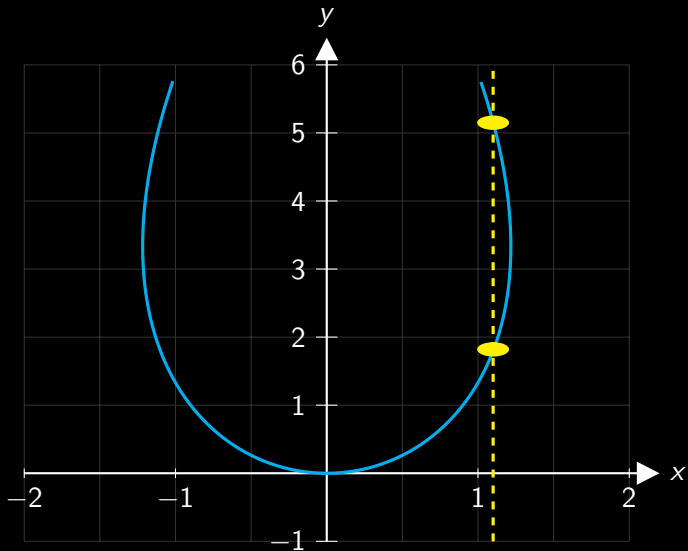


Vertical Line Test

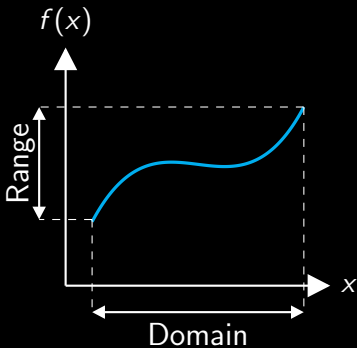
$$x = y^2$$



Vertical Line Test



Domain and Range



The **DOMAIN** of a function is the set of all input values x for which the function is defined.

The **RANGE** of a function is the set of all possible output values $f(x)$ returned by the function.

Domain

- ▶ Is there a value that would cause a divide by zero?
- ▶ Is there a value that would cause the square root of a negative number?
- ▶ Is there a value that would cause the logarithm of a negative number (or zero)?

Domain

► **Divide by zero:** $f(x) = \frac{1}{x-2}$

Domain: $x < 2$ or $x > 2$.

► **Square root of a negative number:** $f(x) = \sqrt{x+3}$

Domain: $x \geq -3$.

► **Logarithm of a negative number or zero:**

$$f(x) = \ln(x-1)$$

Domain: $x > 1$.

Range

- ▶ **Linear function:** $f(x) = 2x + 1$

Range: all real numbers, $-\infty < f(x) < \infty$.

- ▶ **Quadratic function:** $f(x) = x^2 + 3$

Range: $f(x) \geq 3$.

- ▶ **Exponential function:** $f(x) = e^x + 2$

Range: $f(x) > 2$.

Inverse Functions

Given $f(x) = 3x + 1$, find $f^{-1}(x)$.

$$\text{Let } y = 3x + 1$$

$$y - 1 = 3x$$

$$\frac{y - 1}{3} = x$$

$$x = \frac{y - 1}{3}$$

$$f^{-1}(y) = \frac{y - 1}{3}$$

$$\therefore f^{-1}(x) = \frac{x - 1}{3}$$

Inverse Functions

Given $f(x) = 5e^{4x}$, find $f^{-1}(x)$.

$$\text{Let } y = 5e^{4x}$$

$$\frac{y}{5} = e^{4x}$$

$$\ln\left(\frac{y}{5}\right) = 4x$$

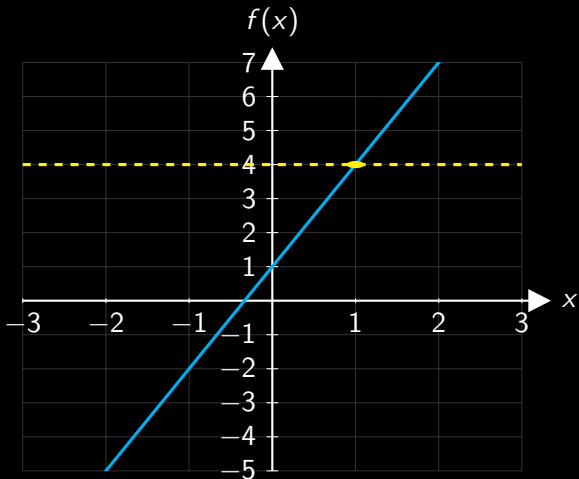
$$x = \frac{1}{4} \ln\left(\frac{y}{5}\right)$$

$$f^{-1}(y) = \frac{1}{4} \ln\left(\frac{y}{5}\right)$$

$$\therefore f^{-1}(x) = \frac{1}{4} \ln\left(\frac{x}{5}\right)$$

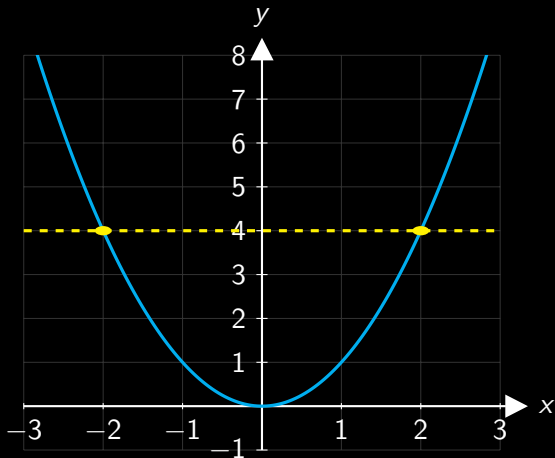
Horizontal Line Test

$$f(x) = 3x + 1$$



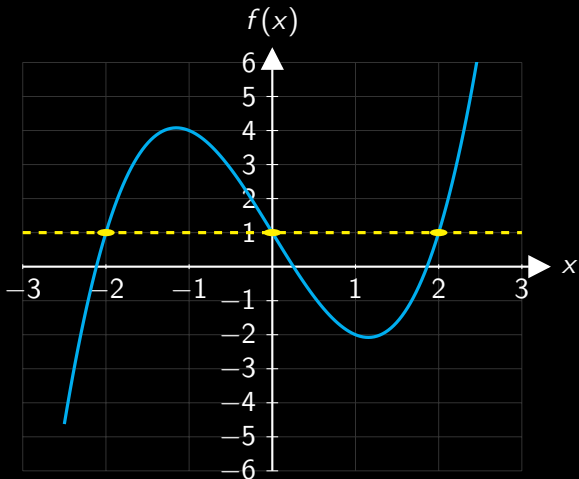
Horizontal Line Test

$$y = x^2$$



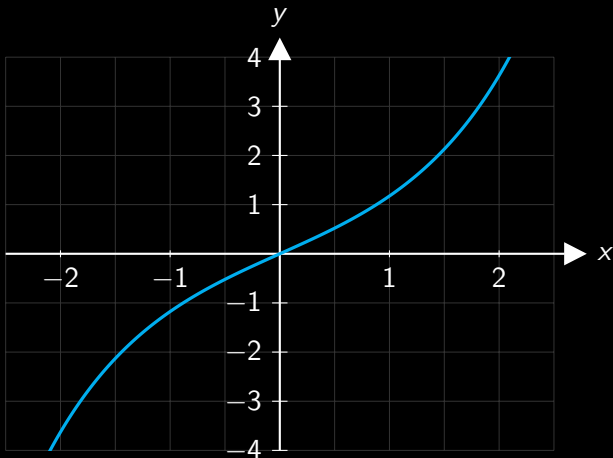
Horizontal Line Test

$$f(x) = x^3 - 4x + 1$$



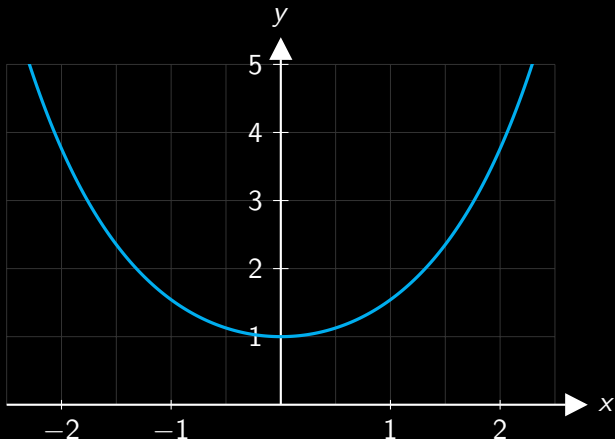
Hyperbolic functions

$$y = \sinh(x) = \frac{e^x - e^{-x}}{2}$$



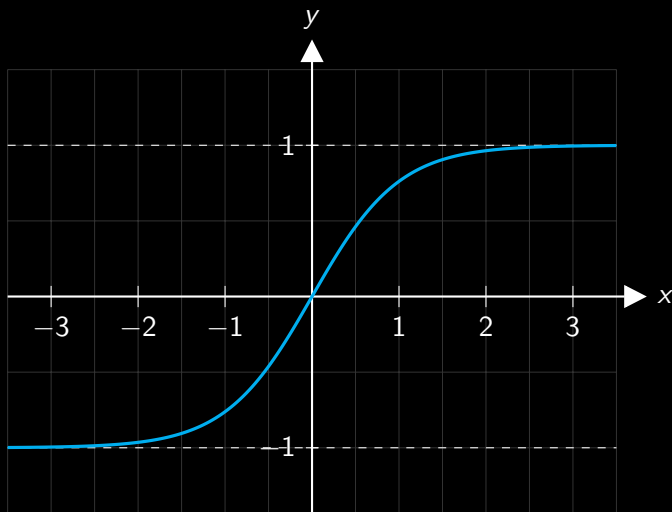
Hyperbolic functions

$$y = \cosh(x) = \frac{e^x + e^{-x}}{2}$$



Hyperbolic functions

$$y = \tanh(x) = \frac{\sinh(x)}{\cosh(x)} = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$



Hyperbolic vs Trigonometric

Hyperbolic

$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$

$$\cosh(x) = \frac{e^x + e^{-x}}{2}$$

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$$

Trigonometric

$$\sin(x) = \frac{e^{ix} - e^{-ix}}{2i}$$

$$\cos(x) = \frac{e^{ix} + e^{-ix}}{2}$$

$$\tan(x) = -i \frac{e^{ix} - e^{-ix}}{e^{ix} + e^{-ix}}$$

Completing the Square

$$y = x^2 + bx + c$$

$$y = x^2 + bx + \left(\frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2 + c$$

$$y = \left(x + \frac{b}{2}\right)^2 - \frac{b^2}{4} + c$$

Completing the Square

$$y = x^2 + 6x + 5$$

$$y = x^2 + 6x + \left(\frac{6}{2}\right)^2 - \left(\frac{6}{2}\right)^2 + 5$$

$$y = x^2 + 6x + 9 - 9 + 5$$

$$y = (x + 3)^2 - 4$$

Completing the Square

$$y = x^2 - 8x + 10$$

$$y = x^2 - 8x + \left(\frac{-8}{2}\right)^2 - \left(\frac{-8}{2}\right)^2 + 10$$

$$y = x^2 - 8x + 16 - 16 + 10$$

$$y = (x - 4)^2 - 6$$

Completing the Square

$$y = x^2 + 4x - 1$$

$$y = x^2 + 4x + \left(\frac{4}{2}\right)^2 - \left(\frac{4}{2}\right)^2 - 1$$

$$y = x^2 + 4x + 4 - 4 - 1$$

$$y = (x + 2)^2 - 5$$

Pascal's Triangle

Row 0					1			
Row 1				1		1		
Row 2			1		2		1	
Row 3		1		3		3		1
Row 4		1	4		6		4	1
Row 5	1		5	10		10	5	1

Binomial Theorem

Example: Expand $(x + y)^3$.

$$(x + y)^3 = 1x^3y^0 + 3x^2y^1 + 3x^1y^2 + 1x^0y^3$$

$$= x^3 + 3x^2y + 3xy^2 + y^3$$

Binomial Theorem

Example: Expand $(x + y)^4$.

$$(x + y)^4 = 1x^4y^0 + 4x^3y^1 + 6x^2y^2 + 4x^1y^3 + 1x^0y^4$$

$$= x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$$

Composite Functions

Please revise composite functions before your Week 2 workshop.

A video is available on [Learning@Griffith](#).

Alternatively, composite functions were covered as part of the Week 8 content in Foundation Mathematics.